YEAR 11 MATHS METHODS INVESTIGATIVE TASK 4 SOLUTIONS

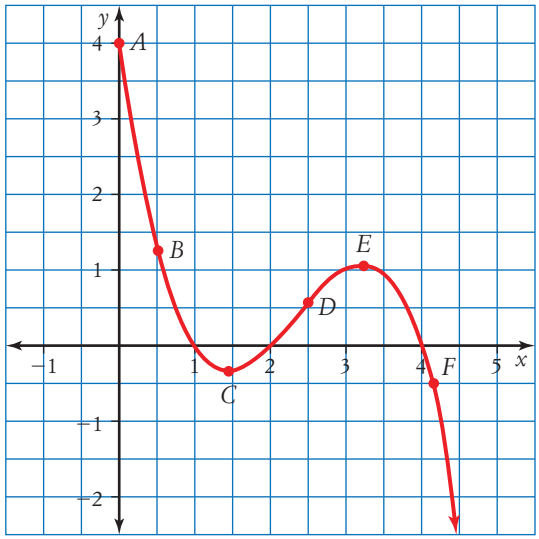
Introduction to differential calculus

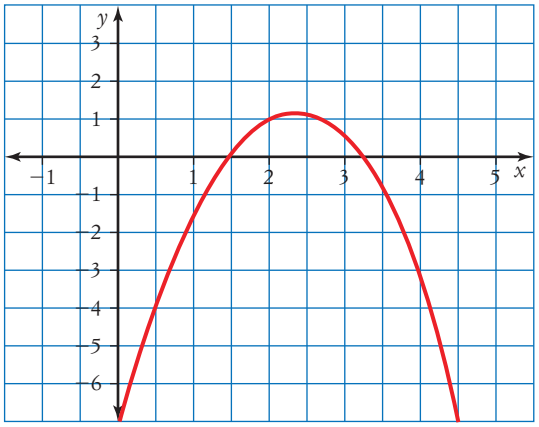
1 a Answers should be very roughly close to these.

$A = -7,\ B = -4\left(-3\frac{7}{8}\right),\ C = 0,\ D = 1\left(1\frac{1}{8}\right),\ E = 0,\ F = -4.5.$

C and E should be exactly 0. **[1 mark each]**

b





[Graph 2 marks, turning points and gradients match 2 marks]

c Explanations will vary but should be similar to the answers which follow, as well as explaining that the gradient does not change sign after an inflection.

[2]

At point C the gradient to the left of the turning point is negative and to the right of the turning point is positive $\therefore$ The point is a minimum. **[2]**

At point E the gradient to the left of the turning point is positive and to the right of the turning point is negative. $\therefore$ The point is a maximum. **[2]**

d Expanding gives $g(x) = x^3 - 7x^2 + 14x - 8$ **[1 mark (can use CAS)]**

$$\begin{aligned} g'(x) &= \lim_{h\to 0} \frac{g(x+h)-g(x)}{h} \\ &= \lim_{h\to 0} \frac{(x+h)^3 - 7(x+h)^2 + 14(x+h) - 8 - (x^3 - 7x^2 + 14x - 8)}{h} \\ &= \lim_{h\to 0} \frac{3x^2h + 3xh^2 + h^3 - 14xh - h^2 + 14h}{h} \\ &= \lim_{h\to 0} \frac{h(3x^2 + 3xh + h^2 - 14x - h + 14)}{h} \\ &= \lim_{h\to 0} 3x^2 + 3xh + h^2 - 14x - h + 14 \\ &= 3x^2 - 14x + 14 \end{aligned}$$ **[1]**

e Using x-intercepts, equation must be $a(x-1)(x-2)(x-4)$ **[1]**

Using y intercept ($x = 0$) gives $a(0-1)(0-2)(0-4) = 4$

Thus $-8a = 4 \Rightarrow a = -0.5$ **[1]**

Thus $f(x) = -0.5(x-1)(x-2)(x-4)$ **[1]**

f Clearly, $f(x) = -0.5g(x)$

The rules for differentiation mean that $f'(x) = -0.5\,g'(x)$ **[1]**

$f'(x) = -0.5(3x^2 - 14x + 14) = -1.5x^2 + 7x - 7$ **[1]**

At A, $x = 0$, $f'(x) = -1.5x^2 + 7x - 7 = -7$ **[1]**

At D, $x = 2.5$, $f'(x) = -1.5 \times 2.5^2 + 7 \times 2.5 - 7 = 1.125$ **[1]**

2 Consider the graph of $y = x^4 - 2x^3 + 1$.

a $\dfrac{dy}{dx} = 4x^3 - 6x^2$ **[2]**

b $\dfrac{dy}{dx} = 0 \Rightarrow 4x^3 - 6x^2 = 0$ **[1]**

$2x^2(2x - 3) = 0 \Rightarrow x = 0$ or $x = 1\frac{1}{2}$ **[1]**

$x = 0 \Rightarrow y = 1$, giving $(0, 1)$ **[1]**

$x = 1\frac{1}{2} \Rightarrow y = \left(1\frac{1}{2}\right)^4 - 2 \times \left(1\frac{1}{2}\right)^3 = -1\frac{11}{16}$, giving $\left(1\frac{1}{2}, -1\frac{11}{16}\right)$ **[1]**

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Draw a table, choosing easy values on either sides of the turning points.

x	-1	0	1	$1\frac{1}{2}$	2
Gradient	-10	0	-1	0	8
	\	—	\	—	/

The first stationary point is a downward inflection and the other is a local minimum. **[3]**

c The dominant term is x^4. **[1]**

Thus as $x \longrightarrow \infty$, $y \longrightarrow \infty$ and as $x \longrightarrow -\infty$, $y \longrightarrow \infty$ **[1]**

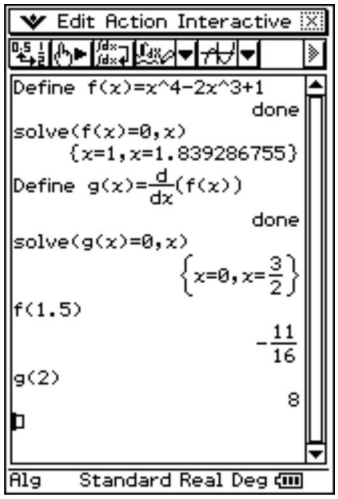
d y-intercept is when $x = 0 \Rightarrow y = 1$, giving $(0, 1)$ **[1]**

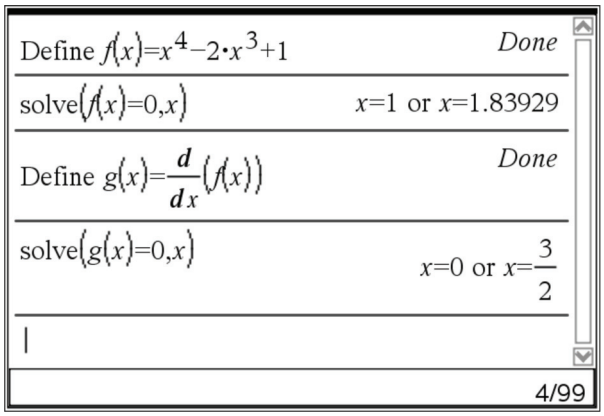
x-intercepts are when $y = 0 \Rightarrow x = 1$ or $x \approx 1.84$ giving $(1, 0)$ and $(1.84, 0)$

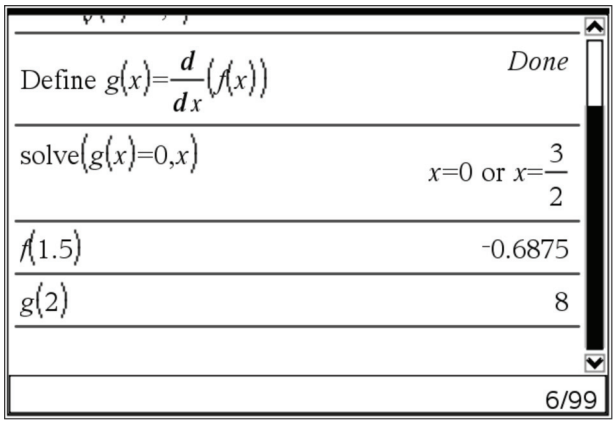
[2]

Drawing the curve (see below) **[3]**

Note: Students could use their CAS calculators for some of these calculations (see below).







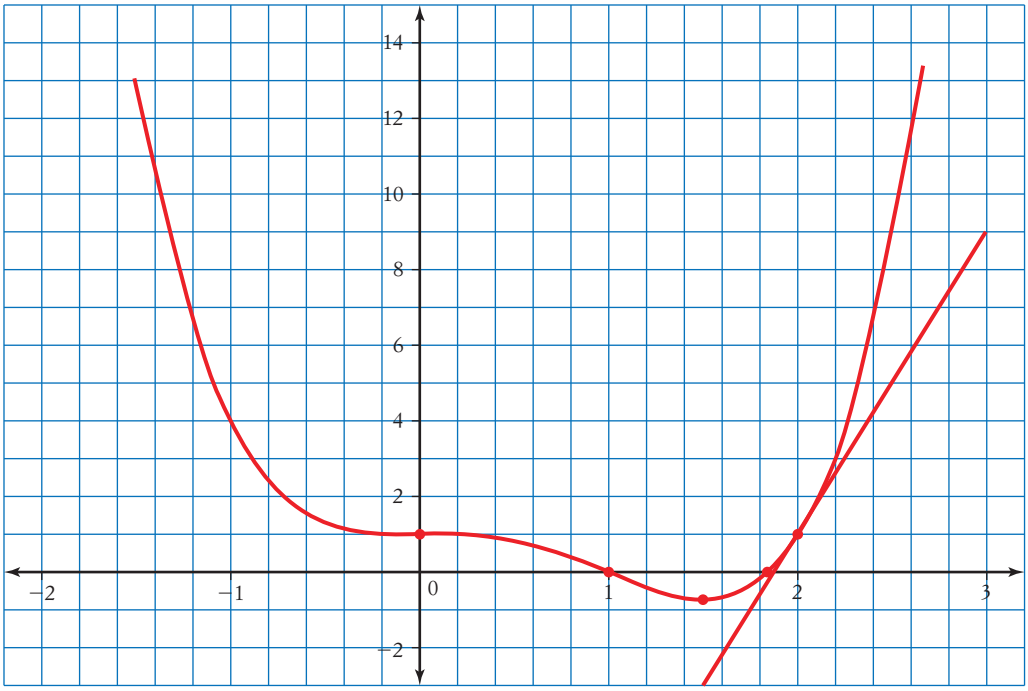
e When $x = 2$, the gradient is 8 and $y = 2^4 - 2 \times 2^3 + 1 = 1$ **[2]**

$y - y_1 = m(x - x_1)$

$y - 1 = 8(x - 2)$

$y = 8x - 15$ **[2]**

Graph for **d**, **e** and **f**



Drawing the tangent:

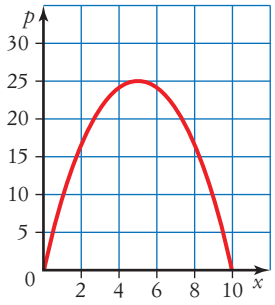
It goes through (2, 1) and gradient 8 means it goes through (3, 9) **[2]**

3 a

x	y	Product
0	10	0
1	9	9
2	8	16
3	7	21
4	6	24
5	5	25
6	4	24
7	3	21
8	2	16
9	1	9
10	0	0

The maximum product is 25. **[2]**

b The product can be expressed as $P = x(10 - x)$



Students may use their CAS calculators for the sketch

The maximum product is 25. **[2]**

c $P = x(10 - x) = 10x - x^2$ **[1]**

$\frac{dP}{dx} = 10 - 2x$ **[1]**

Turning point where $\frac{dP}{dx} = 0 \Rightarrow 10 - 2x = 0 \Rightarrow x = 5$ **[1]**

When $x = 4$, $\frac{dP}{dx} = 2 > 0$ and when $x = 6$, $\frac{dP}{dx} = -2 < 0$

$\therefore$ maximum **[2]**

When $x = 5$, $P = 25$. The maximum product is 25. **[1]**

4 a i 0.5 km/min, 1 km/min. **[2]**

ii $0.5t$ or $\frac{1}{2}t$ km. Counting east (right) as positive, let the blue car's position at time t be x (km) east of the intersection.

Since the car is travelling west, $x = -0.5t$. **[2]**

iii The red car takes 6 min to reach the intersection, travelling at 1 km/min.

The distance travelled in this time is $6 \times 1 = 6$ km. **[1]**

At 11.00 a.m. the red car is 6 km north of the intersection. **[1]**

Counting north as positive, let the red car's position at time t be y (km) north of the intersection.

$v = \frac{dy}{dt} = -1$ **[1]**

$y = -t + c$ **[1]**

When $t = 0$, $y = 6 \Rightarrow c = 6$

$y = -t + 6$ or $y = 6 - t$ **[1]**

iv Using Pythagoras' theorem, $d^2 = x^2 + y^2 = 0.25t^2 + (6 - t)^2$.

$d^2 = 0.25t^2 + 36 - 12t + t^2 = 1.25t^2 - 12t + 36$ **[2]**

b d will be a minimum when d^2 is a minimum.

It may help to use another variable, say $h = d^2$.

$\frac{dh}{dt} = 2.5t - 12$ **[1]**

Find $\frac{dh}{dt} = 0 \Rightarrow 2.5t - 12 = 0 \Rightarrow t = 4.8$ **[1]**

The time is 4.8 min. = 4 min. 48 s. **[1]**

c It is probably easiest to use $d^2 = 0.25t^2 + (t-6)^2$ and t = 4.8.

$d^2 = 0.25 \times 4.8^2 + (-1.2)^2 = 7.2$ **[1]**

$d = \sqrt{7.2} = \frac{\sqrt{72}}{\sqrt{10}} = \frac{\sqrt{720}}{\sqrt{100}} = \frac{12\sqrt{5}}{10} = \frac{6\sqrt{5}}{5}$ (exact) **[2]**

Note: At this level students should be able to simplify surds to the best form.

$d = \sqrt{7.2} \approx 2.68$ **[1]**

The cars' smallest distance apart is $\frac{6\sqrt{5}}{5}$ km which is about 2.68 km.

Alternative method for parts (ii) and (iii)

ii $0.5t$ or $\frac{1}{2}t$ km. Counting west (left) as positive, let the blue car's position at time t be x (km) south of the intersection. Then $x = 0.5t$. **[2]**

iii The red car takes 6 min to reach the intersection, travelling at 1 km/min.

The distance travelled in this time is $6 \times 1 = 6$ km. **[1]**

At 11.00 a.m. the red car is 6 km north of the intersection. **[1]**

Count south as positive, and let the red car's position at time t be y (km) south of the intersection.

$v = \frac{dy}{dt} = 1$ **[1]**

$y = t + c$ **[1]**

When $t = 0$, $y = -6 \Rightarrow c = -6$

$y = t - 6$ **[1]**

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